[MOP] PCD-V onboarding for the SA team

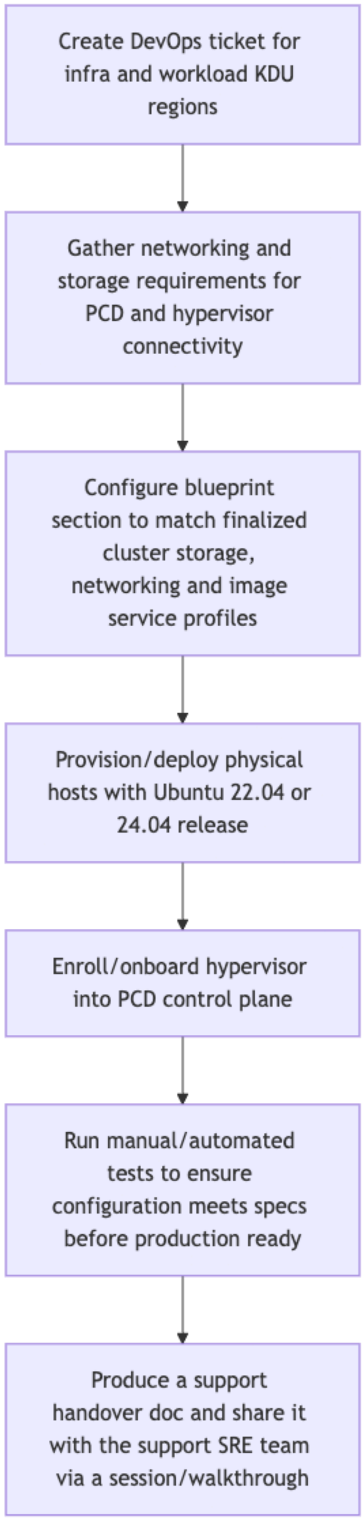
🏹 Objectives

- Defining and capture working procedures for setting up and configuring PCD specific to customers environment.

> ⚠️ At the beginning of the engagement calls with the customer ensure a directory with the customer name is created under the customer_success google drive

Setting up and configuring PCD -V

General flow of tasks



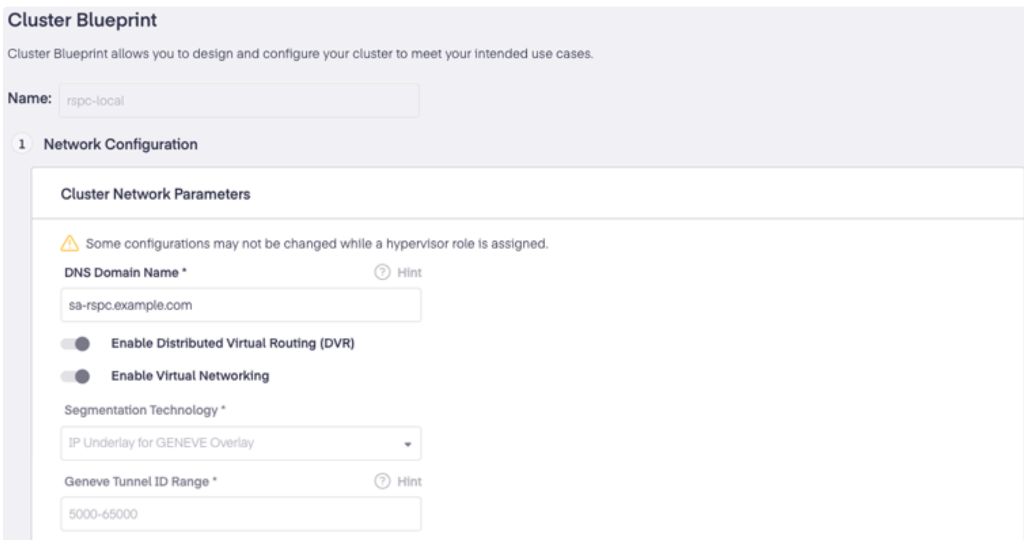
NOTE: Devops ticket is **not needed** for self-hosted environment deployments. [Follow the self-hosted deployment guide] to setup the control plane in the customer's environment.

- During the discussion, gather details regarding the preferred KDU region names and infra region short name from the customer.
- Open a devops request with the names of the region that is to be created in the production SaaS plane.

Infrastructure configuration in the cluster blueprint page

This section assumes that the discovery and understanding of customers infrastructure is complete and all the necessary information is ready to be set in the **cluster blueprint configuration** section of a region

Cluster creation



- The Name field defines the name of PCD cluster for the whole region. Once set it cannot be changed.
- The Network configuration section will primarily define the overlay network layout that will be used for the virtual machine traffic.

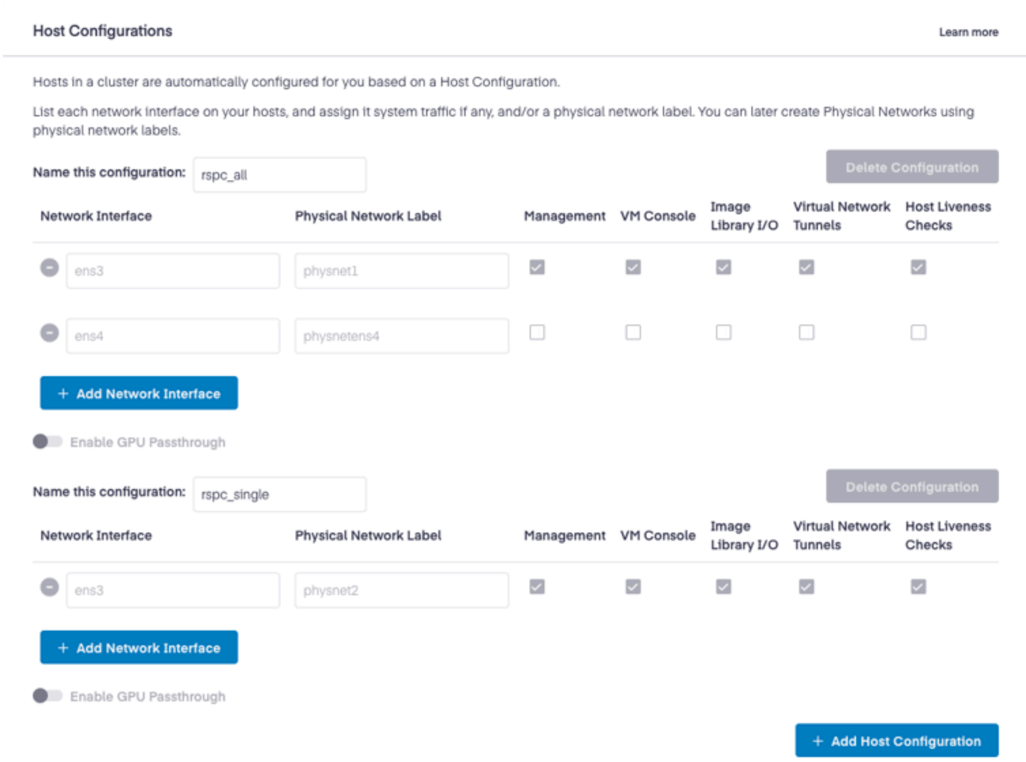
> **Key items to keep in mind**
>
> - The DNS domain Name should ideally be the one that is shared by the customer. This will be used for setting up FQDN for the VMs (hostnames)
> - The Domain name set in the blueprint still does not guarantee that it will be used during VM FQDN creation. It will work only when coupled with DNS service like designate.
> - For external DNS/IPAM integration, the FQDN names from external service can still take precedence over this value.

🛟 Resources

- [cluster blueprint documentation]

Networking

From the cluster blueprints page configure the host configurations section as per the customer network make up



In a typical datacenter environment the networking infrastructure would be VLAN segregated. During discovery calls and other engagement with the customer, ensure to get the following details from the customer’s team:

- A layout diagram of the network used for the current setup (Mostly VMware ones)
- Actual application/workload requirements that is getting fulfilled in the existing layout.
- DHCP/DNS/IPAM implemented in the present architecture.

Things to remember when setting up networking in the blueprint section

- The interface defined in this within the network profile will be used by PCD for communication (service and VM connections alike).
- For interface to be considered and mapped within PCD, it would **either need to have a label set or it should have one of the PCD traffic mapped on to it**.
- Interfaces not part of this configuration section, **can continue to exists** and be used for other communications by the hosts while being part of the PCD cluster.

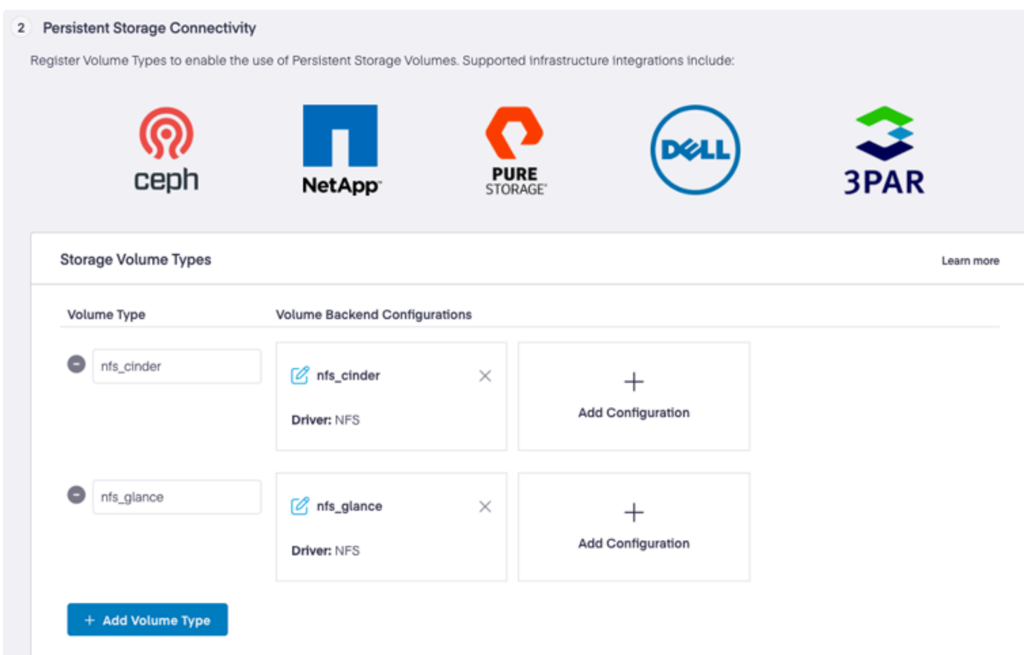
Limitation with openstack neutron service:

- Openstack neutron **will always assume itself authoritative** for IPAM/DNS services
- As of writing, in the current release, Neutron will not directly integrated with any external DHCP/DNS providers without any specific workarounds

Resources

- [SaaS hosted PCD deployment guide]

Storage



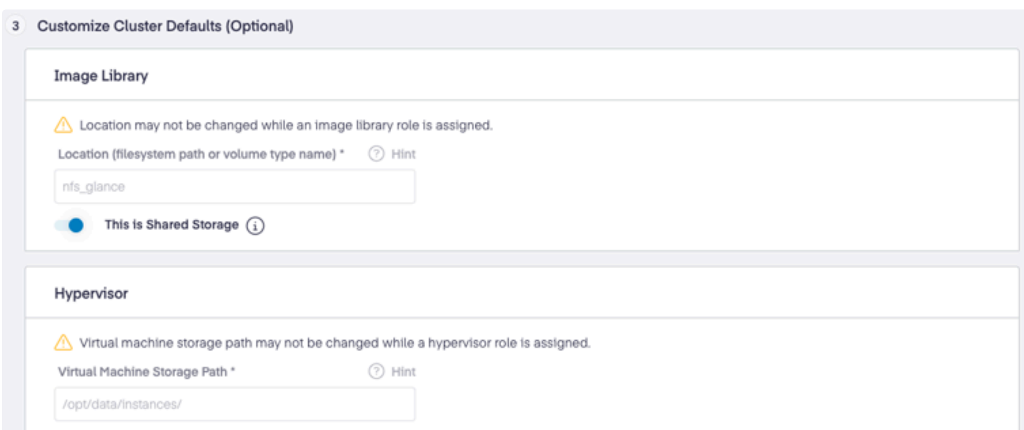
- The `volume_type`, as the name suggest, defines the type of the storage.
- The `Volume backend section` is to be used to define the actual storage driver configuration

🛟 Resources

- [List of volume backend configuration that are tested and working in the production PCD environment for various customers]()
- [Block storage configuration doc]()

> **Things to remember when setting up Storage in the blueprint section**
>
> - Multiple hosts in a cluster can have the storage role/profile applied.
> - Setting up multiple backend configuration for the same volume type will setup cinder to load balance and use all the defined backend within that volume type.
> - Any changes in the backend configuration will get automatically pushed to the storage hosts in the cluster without any service restarts etc.
> - **Cinder service is applicable for the whole region. Hence every hosts in a region is expected to have a view/access to the storage backend that is served by the storage host/hosts**

Image service and ephemeral storage



- In the current form Glance/image library backend can either be cinder backed or external managed.
  - To setup cinder backed configuration enter the `volume_type` name in the given field.
  - For externally managed storage configuration provide the mount directory name. This directory should be present and the backend is to be mapped to it on the respective glance role host.

> **Things to remember when setting up Image library in the blueprint section**
>
> - Glance/image library role is applied per region.

- Multiple hosts in the cluster can have glance role applied for an HA like functionality
- For externally managed shared store, PCD will not manage them. In those case the respective mount point and backend should be created and managed outside of PCD manually or via automation.
- There is an upload restriction of upto 5GB for the images via the PCD portal UI.
- For large sized images the command line method such as should be followed:

```
image create --os-interface public  --insecure --container-format bare --disk-format qcow2  --private --file ~/Downloads/cirros-0.6.3-x86_64-disk.img Cirros_test
```

- The `hypervisor` section can be used to define the ephemeral storage directory if needed else this will default to the provided one which will be local to the hypervisor host only.

Resources

- PCD image library documentation
- remote site integration into single region (Special Use case)

Automation to setup/configure region blueprints

Provisioning and onboarding physical hosts to PCD

Hardware:

https://platform9.com/docs/private-cloud-director/private-cloud-director/pre-requisites

Pre-requisites

This should be performed by the customer in coordination with their network/datacenter groups accordingly: PCD-V hardware onboard Pre-requisites

Networking:

Types of network architecture typically expected in the Datacenter hosts

Onboarding to PCD

This is a two step operation:

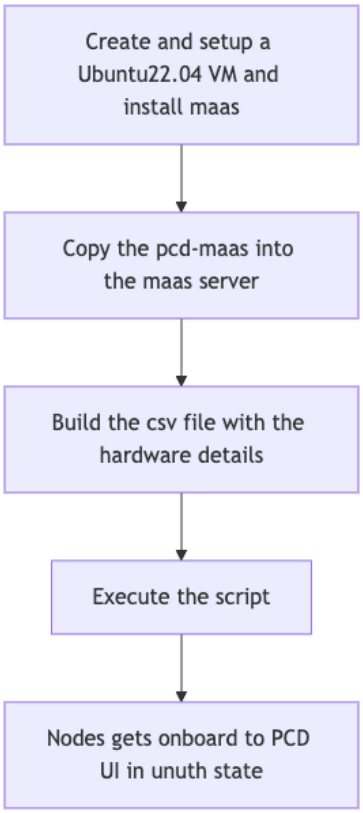
- Run the pcdctl tool on each of the hosts to be enrolled into the pcd cluster.
- Assign the required roles to the hosts visible in the UI under the cluster hosts section.

Things to remember

- The host management operation like authorising/de-authorising from the UI can only be performed by user with **Admin** role assigned

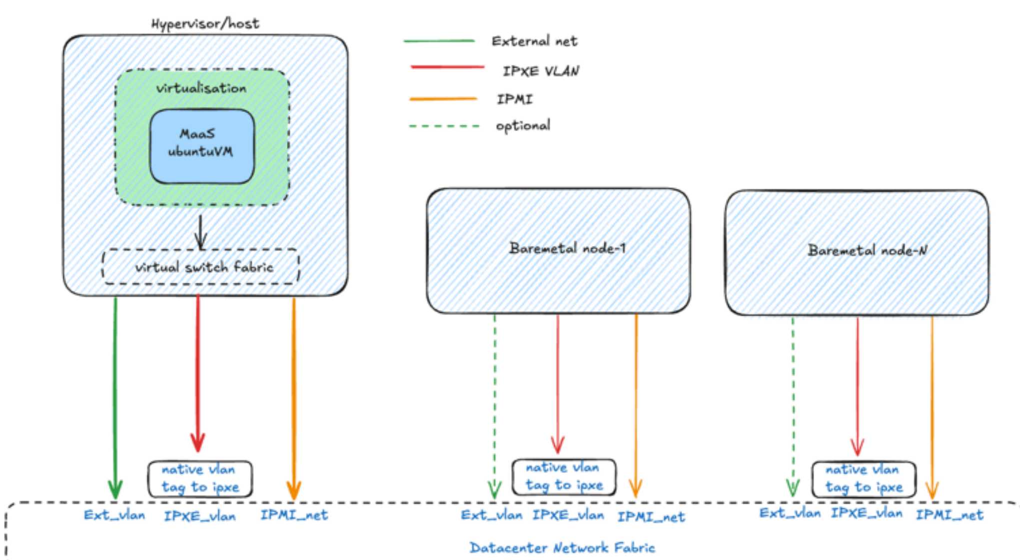
Automation to onboard hosts to PCD

For onboarding the baremetal hosts via a scripted method the process flow would follow:



Network connection requirements

The MaaS server (deployed as a VM or baremetal) requires access to specific network to be able to successfully re-image the host with ubuntu release. The following is a typical network connections for MaaS:



Once this configuration is in place, proceed to next step.

Procedure

- Deploy an Ubuntu 22.04 as a VM which can reside in the VMwmare or on the PCD hypervisor.
- Download/copy the https://github.com/Platform9-Community/sa-automation/blob/main/Maas_installer_script/maas-configure.sh script onto VM.
- Set the execute permission and run the script with `sudo` to install the maas server:

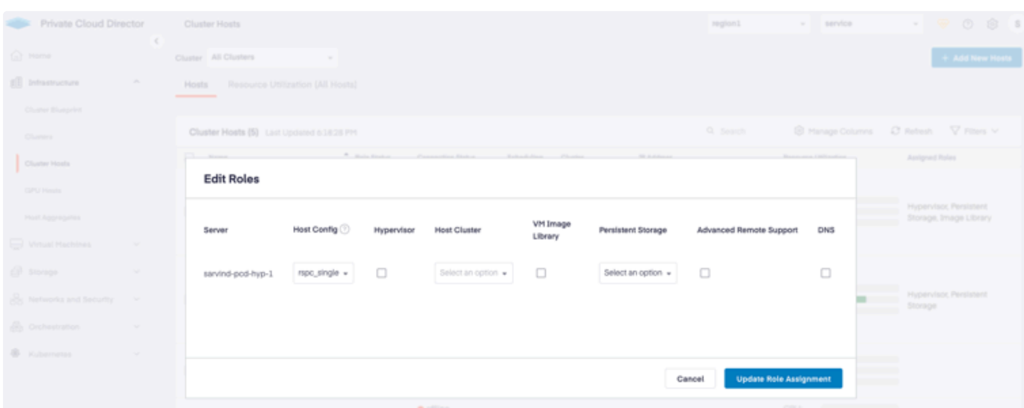
```
sudo chmod a+x maas-configure.sh
sudo ./maas-configure.sh
```

- Once installed, the `MaaS` UI should be reachable over `http://VM-IP:5240/`
- Copy/download the latest pcd-maas script package into the maas server from : https://github.com/platform9/pcd-maas/tags and extract it locally.
- After extraction follow the steps provided in the repo readme file: https://github.com/platform9/pcd-maas
- After the execution of the script the set of target hosts will now be visible in `un-authorised` state in the PCD UI.

Applying the required roles to the hosts

Portal UI method

Depending upon the requirements to make use of hosts, it would need to be authorised by assigning specific roles to them which can be done by selecting the hosts and the needed role in the cluster hosts page:



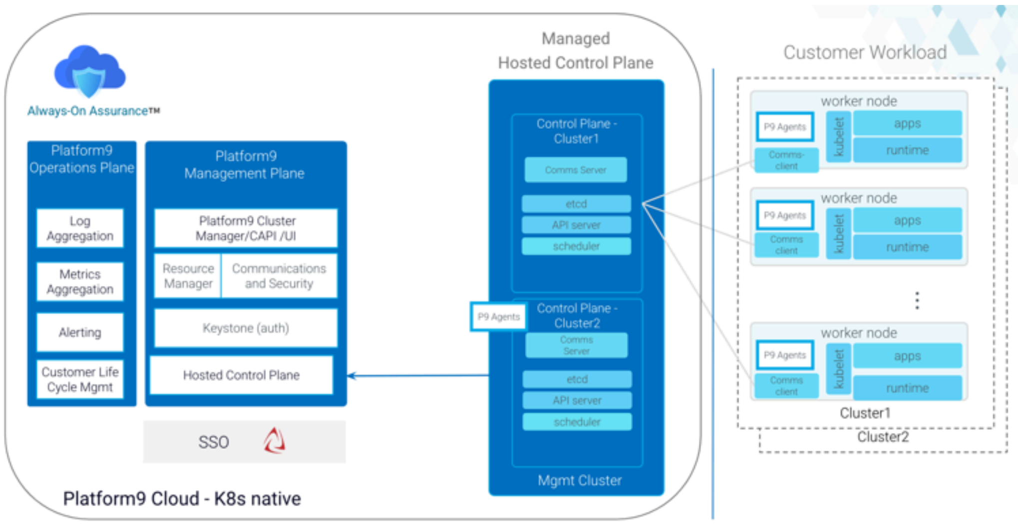
Automated way:

- Roles can be assigned in batches using scripted way via the pcdexpressv2 framework: https://github.com/platform9/pcd_ansible/tree/pcdExpressV2

PCD-K Deployment Consideration

- Once PCD setup is done, based on the engagement requirement if Customer requires an orchestration platform, then a PCD Kubernetes platform is deployed where the Control plane is hosted on the EKS AWS side.

- The customer is responsible to manage the applications running on the workload cluster. As of this writing, Self hosted Control plane is not enabled on PCD side.

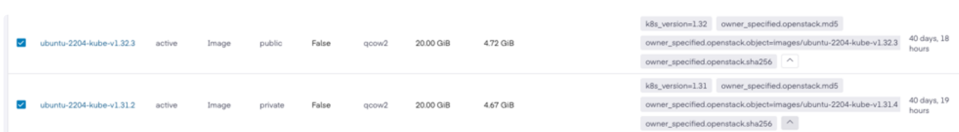


🛠️ **Set-up Prequisites**

- [Pre-Req documentation]
- Cluster-API k8s images should be available in Glance Registry before creating Kubernetes Workload Cluster. The images need to be fetched from **[quay]** repo.
- To convert & upload the k8s images on the Glance refer to:

  [Fetch k8s vm images from quay repository to local machine for PCD]

- Once uploaded, the images should be tagged in the form ***k8s_version***=<***major.minor***> e.g - k8s_version=1.32.



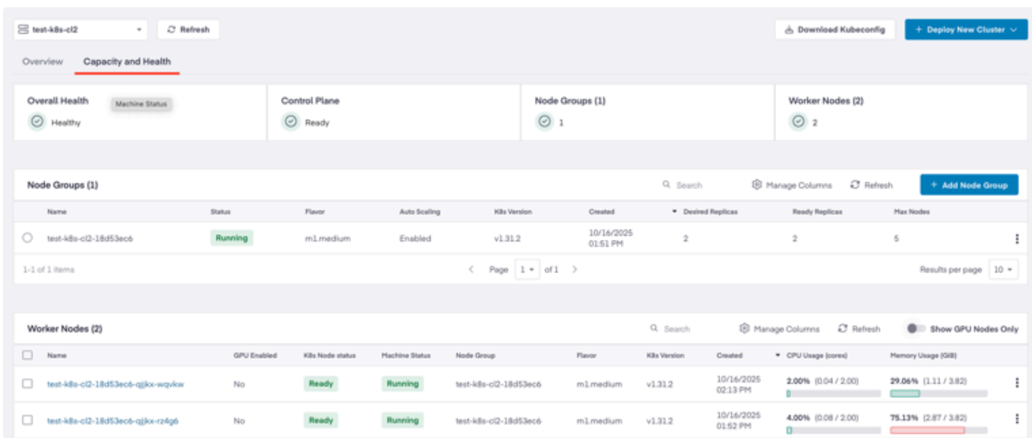
☑	ubuntu-2204-kube-v1.32.3	active	Image	public	False	qcow2	20.00 GiB	4.72 GiB	k8s_version=1.32 owner_specified.openstack.md5 owner_specified.openstack.object=images/ubuntu-2204-kube-v1.32.3 owner_specified.openstack.sha256	40 days, 18 hours
☑	ubuntu-2204-kube-v1.31.2	active	Image	private	False	qcow2	20.00 GiB	4.67 GiB	k8s_version=1.31 owner_specified.openstack.md5 owner_specified.openstack.object=images/ubuntu-2204-kube-v1.31.4 owner_specified.openstack.sha256	40 days, 19 hours

- (Optional) Ensure that the Control plane pods in the **Kaapi** namespace in the AWS Environment are in Running state before triggering PCD-K deployment

```
➜  ~ k get pods -n kaapi
NAME                                                      READY   STATUS    RESTARTS       AGE
access-manager-56bc566f7-54z5t                            1/1     Running   3 (5d2h ago)   51d
addon-controller-ccf6dc8bf-rmmjg                          1/1     Running   2 (5d2h ago)   41d
barista-controller-manager-76d7d87cdc-rq8n4               1/1     Running   6 (5d2h ago)   41d
byoh-controller-manager-5585ff9d78-njrbm                  2/2     Running   0              46h
classifier-manager-b844bf558-w6sbh                        1/1     Running   23 (45h ago)   41d
event-manager-7cf6c4ffdc-ttqbt                            1/1     Running   3 (5d2h ago)   41d
hc-manager-7ff784dccb-79sm6                               1/1     Running   2 (5d2h ago)   51d
hcp-controller-manager-fd744ff47-br7z4                    1/1     Running   5 (5d2h ago)   41d
kaapi-management-plane-cluster-api-operator-8c5488998-z22cs   1/1     Running   0              43h
kaapi-management-plane-kamaji-5b54649b56-k29ts            1/1     Running   0              43h
kaapi-management-plane-kube-state-metrics-555cf85848-ghbjs   1/1     Running   2 (5d2h ago)   41d
kamaji-etcd-0                                             1/1     Running   0              41d
kamaji-etcd-1                                             1/1     Running   0              42h
kamaji-etcd-2                                             1/1     Running   0              41d
kamaji-etcd-client                                        1/1     Running   0              50d
sc-manager-7b6b8ccb9-q4s58                                1/1     Running   2 (5d2h ago)   41d
shard-controller-787f69979-cstrt                          1/1     Running   0              51d
```

🚀 **PCD-K Deployment & Cluster-access Steps**

Procedure - https://platform9.com/docs/private-cloud-director/private-cloud-director/getting-started-with-kubernetes-in-pcd



Accessing the cluster - https://platform9.com/docs/private-cloud-director/private-cloud-director/kubernetes-cluster-access-kubeconfig

Post setup procedures for the region handover

Sanity testing

- Before handing over the region to customer to deploy production workload, make a copy of Sanity-test-template, into the customer specific directory in the google drive.
- Run those tests in the sheet and mark its status accordingly to track any of the issues noted.
- Actively try to troubleshoot the issue and engage the Engineering team if necessary to fix the issue.
- Raise an bug/RFE Jira if needed **after properly identifying the issue being a shortcoming in the PCD framework.**

Support handover process

- Post the region released to production for the customer after sanity testing, create a support handover doc with the customer name under the: Customer Overviews location.
- Create the doc using the template that will be available under the template section during adding and editing the page.
- Prepare and send an invite to the SRE support team members for a session to elaborate on the deployments and configuration done for the customer

Migrating virtual machines from VMware to PCD environment with Vjailbreak

- Vjailbreak is gets deployed as a VM on PCD hypervisors only.
- The VM need to be deployed in a network has access to VMware as well the pcd cluster.
- To deploy and configure Vjailbreak refer to : https://platform9.github.io/vjailbreak/introduction/getting_started/

Resources

Resource	Link
Slack channel for PCD-V	#cloud-director
Slack channel for Vjailbreak tool	#vjailbreak
Slack channel for PCD-k framework	#cloud-director-kaapi
SA team channel	#solutions-consulting
PCD-V useful info landing page	PCD Useful Info
PCD-K useful info landing page	https://platform9.atlassian.net/wiki/spaces/OP/pages/4458250252/PCD+Useful+Info